

Clarissa Röder *Software Engineer*

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📁 Experience

CERN (European Organisation for Nuclear Research), *Software Engineer* Apr 2025 – Present | Geneva

- Developed and maintained full-stack applications for CERN's Product Lifecycle Management (PLM) platform, supporting over 5,000 engineers and technicians managing design data and technical documentation for the accelerator complex
- Took primary responsibility for implementing the opening feature of the Emergency Open and Save initiative, delivering critical functionality under tight deadlines for high-priority engineering operations
- Built the Change Authoring Tool feature, enabling users to switch the CAD authoring tool of documents with automatic versioning and native file handling, implementing both the React/TypeScript UI and C# Aras backend
- Configured and enhanced the EBOM Export feature, adding profile card access and fixing calculation bugs including incorrect total quantities across nested BOMs
- Led the React 19 upgrade, managing dependencies and ensuring application stability
- Contributed to release management, coordinating deployments, tracking merge requests, and managing the testing process for production releases
- Worked closely with business analysts to translate requirements into technical implementations
- Participated in code reviews, testing, and system maintenance using daily standups and Kanban workflows

Flytrap, *Software Engineer* Aug 2024 – Dec 2024

- Built a self-hosted error monitoring platform focused on ease-of-use and data ownership, offering SDKs for seamless integration across four platforms
- Engineered a scalable AWS cloud architecture leveraging API Gateway, SQS, Lambda, RDS, EC2, S3, and SNS
- Streamlined infrastructure deployment with Terraform, abstracting the creation of 60+ AWS resources into 5 steps
- Designed and implemented a RESTful API using Flask and Docker, with a React/Chakra UI dashboard featuring advanced filtering and interactive stack trace views
- Implemented secure source map management, automated stack trace unminification, and near real-time alerts using SNS and WebSockets
- Collaborated remotely with a global team of engineers across three time zones using agile workflows and Git best practices

📁 Projects

- 🌲 Advent of Code: Solved 100+ complex algorithm challenges, enhancing problem-solving and coding proficiency
- RequestHub: Built a webhook inspection and debugging tool using React, Express, Node.js, MongoDB, PostgreSQL, WebSockets, NGINX, and DigitalOcean
- QuickSplit: Developed a shared expense tracking application using Ruby, ERB, and Sinatra
- Margit Lisa Photography: Developed a responsive Next.js website to showcase high-resolution fine art photography

📁 Skills

Languages & Frameworks	Cloud & Infrastructure	Other
JavaScript, TypeScript, C#/.NET, Python, Ruby, SQL, React, Express, Flask, Next.js, Three.js, Electron	AWS (EC2, Lambda, API Gateway, S3, RDS, SQS, SNS), Terraform, Docker, DigitalOcean, Heroku	PostgreSQL, MongoDB, SQL Server, Git/GitHub, REST, SOAP, WebSocket, SSE, Logstash, OpenSearch, NGINX, TensorFlow

🎓 Education

LaunchSchool Multi-year, mastery-based software engineering curriculum. Read more at launchschool.com/employers	2023 – 2024
Goldsmiths, University of London , <i>M.Sc. Forensic Psychology</i> With Merit	2017 – 2018
Maastricht University , <i>B.Sc. Psychology</i> Honours Programme, Cum Laude	2014 – 2017

📁 Certificates

CS50xAI Introduction to Artificial Intelligence , <i>HarvardX</i>	2025
CS50x Introduction to Computer Science , <i>HarvardX</i>	2024